logisth 0.9.2 — User Manual



wavelogik logisth is a two-part polyphonic subtractive synthesizer for VST3, CLAP and standalone use. This manual describes logisth's own controls and behaviour. logisth is an independent instrument and is not affiliated with LennarDigital. It does not load Sylenth1 presets or reproduce its sound exactly.

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1. Installation and first sound

Download the archive for your operating system from the GitHub release. Each native archive contains VST3, CLAP and Standalone folders. Extract it first. Copy the entire .vst3 bundle and the .clap file or bundle into a plug-in directory scanned by your DAW. Keep the internal structure of bundles intact.

System	User VST3 directory	User CLAP directory
Windows	%LOCALAPPDATA%\Programs\Common\VST3	%LOCALAPPDATA%\Programs\Common\CLAP
macOS	~/Library/Audio/Plug-Ins/VST3	~/Library/Audio/Plug-Ins/CLAP
Linux	~/.vst3	~/.clap

If your DAW does not scan these user directories, add them to its scan paths or use its documented system directory. Rescan plug-ins, insert logisth on a MIDI instrument track and play a note. A fresh instance automatically selects Init / saw: one classic saw oscillator, both filters bypassed, no effects, arp or modulation, and Main at 65%. Saved presets/projects restore their stored Main value rather than replacing it with the new-instance default. The on-screen keyboard also generates notes. Start at a moderate listening level.

For Standalone, launch the executable or app and choose audio and MIDI devices in its audio settings. The standalone wrapper provides its own device settings; those are separate from the synth panel.

Windows builds target x64. macOS Universal bundles contain Intel x86_64 and Apple Silicon arm64 code and require macOS 11 or later. The minimum Linux glibc version is recorded in BUILD_INFO.json. macOS bundles have an ad-hoc signature, without Apple notarization; Windows binaries have no publisher signature. Follow your OS's normal review/open procedure for downloaded applications you trust.

VST2 is not included: building it requires a separately licensed VST2 SDK.

2. The single-window console

All sections occupy one console. No page needs vertical scrolling. Select A or B at the top to edit that part's two oscillators, filter and amplitude envelope. This selects the editing view; it does not mute the other part. Use SOLO for listening to one part. The central rack changes its controls when you select a rack name. The small square beside each name independently enables that section. Selecting a rack page does not enable or disable its processing.

The upper row contains the preset selector, previous/next preset buttons, A/B, SOLO, POLY, SYNC and SETTINGS. Filters are on the left, the rack in the centre, and the mixer on the right. Mod Env, LFO and Misc routing remain visible below them. The bottom row contains wheels, the keyboard and mono/portamento controls.

Drag the bottom-right resize handle, or open SETTINGS / APPEARANCE and enter UI SCALE from 80% to 200%. Decimal values such as 137.5% are accepted. The nominal size is 1240 by 860 pixels. The displayed percentage shows the last valid scale; invalid/out-of-range entries restore it. Exact window dimensions and the entered percentage are saved and restored when the editor or a new instance opens. A fresh installation adapts the initial size to its display. Each open window keeps its own size; changing one sets the default for subsequent windows.

SETTINGS has APPEARANCE, THEME EDITOR, STYLE and ABOUT tabs inside the central rack. The standard themes are Light, Midnight Blue (default), Cobalt, Graphite and Forest. Theme selection is shared by open instances and recalled by new ones; audio presets/projects preserve theme and scale. Preferences live in `logisth/appearance.json` under the system user application-data directory.

APPEARANCE / OPEN FOLDER reveals the current theme directory. USE FOLDER selects another directory; missing standard themes are created there. EDIT / NEW opens THEME EDITOR. Enter a name and select one of twelve colour roles, then use the colour picker or a six-digit RGB hex colour such as #62ACFF and press Enter. DARK CONTROLS selects light/dark control styling. SAVE NEW writes a custom copy in the current folder. SAVE AS... lets you choose a folder and exact JSON file name. RENAME renames the selected custom theme, including its file; standard files remain protected. RESET restores the selected saved theme.

STYLE selects a font family installed on this computer (Open Sans is embedded), Flat/Gradient/Grid background and Arc/Solid/Minimal knobs. Knob and Knob Track colours are separate from the accent. These choices preview immediately and are saved with the theme. Unsaved edits stay with this instrument instance when its editor closes/reopens or is minimized; save a theme to retain it across sessions. Choosing another saved theme or RESET discards the temporary draft.

Copy shared .json files into the selected folder to install them. The folder is polled every two minutes; RESCAN refreshes it immediately. New files do not change the current selection. Invalid files are ignored; removing the selected file falls back to Midnight Blue. Version-2 themes contain the name, dark flag, twelve RGB colours, font and background/knob styles. Version-1 themes remain readable. Use assets/themes as templates; docs/THEMES.md documents the format.

The editable comment box below SOUND LIBRARY belongs to the sound. Type a note there, then save the user preset or DAW project. Comments support Unicode and up to 4096 characters. Loading a preset replaces its comment; Init starts with an empty comment. The About section lists the current version and framework credits.

Drag knobs and faders to adjust values. Numeric fields can be dragged or edited. Examples of accepted values include 2.5k for 2500 Hz, 350ms and 65%. Double-click a control to restore its default; modulation amount controls return to zero. Hover for a tooltip and the parameter/value readout at the bottom. PHASE remains editable in Classic and Wavetable with RETRIG either off or on.

3. Oscillators and amplitude envelopes

There are four oscillators: A1 and A2 feed source A; B1 and B2 feed source B. Each oscillator has a waveform menu, 0–8 unison voices, volume, octave, note, fine tuning, phase, detune, stereo width, pan, retrigger and invert.

Control	Purpose
Wave	Selects the oscillator waveform.
Voices	Sets the unison voice count. Zero silences this oscillator.
Volume	Sets its level before filtering and the part envelope.
Octave / Note / Fine	Coarse octave, semitone and cent tuning.
Phase / Retrig	Phase is a live 0–360-degree offset. Retrig gives fixed note starts; off keeps random starts while the Phase offset remains active.
Detune	0–100 cents of unison spread. At 100 ct the outer voices are one semitone below/above the note. With one voice, detune has no effect.
Stereo	Spreads unison voices across the stereo field.
Pan	Positions the oscillator output left or right.
Invert	Inverts oscillator polarity; it can affect cancellation when sources mix.

PHASE applies to all unison voices in the selected oscillator and can be changed while notes are held or automated by the host. The offset is independent of RETRIG: with RETRIG on, each new note starts at the chosen phase; with it off, new notes keep randomized base phases and the knob offsets those phases. Toggling RETRIG does not restart a held oscillator. Phase is relative timing, so its audible effect is clearest when mixing periodic oscillators or moving it during a note. Noise is nonperiodic, and Sampler/Granular use sample-position controls instead.

Live knob changes use approximately 5 ms of cyclic smoothing. Zero and 360 degrees are equivalent; 359 to 1 degrees crosses the boundary by two degrees. Existing phase modulation is added to the knob offset and retains its own modulation rate.

The amplitude envelope for each part has attack, decay, sustain and release. Attack rises from silence, decay approaches sustain, and release fades after note-off. Sustain is a level; the other three are times. The line above the

faders illustrates the current envelope. Amp Env A and B remain independent. Zero-time attack and decay enter sustain without emitting an extra peak sample. With A, D and S at zero, the note is silent. Amplitude envelopes use a short, approximately 1 ms anti-click smoothing window, so zero Attack/Release avoid an abrupt gate at an arbitrary waveform phase. This also smooths other amplitude-envelope transitions and adds a bounded tail of up to approximately 1 ms. The settled level is unchanged; modulation envelopes retain their unsmoothed control values. The preview uses a proportional time axis, with a 0.5-second illustrative held segment. Zero-millisecond attack draws a vertical rise; a larger attack time makes the rise take longer. Zero decay/release also draw vertical transitions. There is no artificial minimum width added to these segments.

3.1 Four source engines

Every oscillator slot A1/A2/B1/B2 has its own engine menu: Classic, Wavetable, Sampler or Granular. All four may use different engines and different audio files. The adjacent view menu shows only pages used by the selected engine. Classic has VOICE, ENV and MOD; Wavetable adds WAVETABLE, and Sampler adds SOURCE and PLAYBACK. All engines have MOD, with only the applicable controls. Granular also has GRAINS; only the Granulizer model exposes MOTION and TIME. VOICE holds the shared tuning, level, unison, stereo and pan controls. SOURCE holds the engine-specific controls; ENV holds that oscillator's optional ADSR. The central Amp Env A/B remains the final envelope for its part.

Classic offers Sine, Saw, Pulse, Triangle, White noise, Half pulse, Quarter pulse, Tri-saw, Pink noise, Brown noise, Organ and Folded sine. Pink and Brown have progressively stronger low-frequency spectra. Organ adds octave harmonics; Folded sine applies sinusoidal waveshaping. Noise has no fixed periodic phase.

MOD exposes FM INDEX (0–8 radians) and FM RATIO (0.25–16) for Classic/Wavetable. This is sinusoidal phase modulation, producing FM sidebands; ratio is relative to the carrier pitch. AM DEPTH (0–100%) and AM RATIO (0.125–16) work in all four engines using a unipolar sinusoidal amplitude modulator. Pulse, Half pulse and Quarter pulse also expose WIDTH, PWM DEPTH and PWM RATE (0.01–20 Hz). WIDTH sets the base duty cycle, PWM adds a sinusoidal width sweep. Half/Quarter pulse retain their narrower base widths. Other waves hide these pulse controls. Manual modulation controls use a short smoothing ramp; ratios are per player. Wavetable treats the file as a bank of consecutive, single-cycle waves. **WT POS** selects one frame; the selected cycle repeats at the played note's pitch. It does not play the entire file in sequence, and adjacent frames are not blended. Positions 0% and 100% select the first and last frames. Every frame is reachable; a 128-frame file provides 128 selectable waves. WARP bends phase, and precomputed spectral bands reduce aliasing at higher notes. Unison voices share one position smoother, so adding voices does not change the speed of WT POS movement.

Click the oscillator's waveform strip to open the central source window. In Wavetable mode, it shows a perspective stack of waves, highlights the current frame and displays its one-based number and total frame count. While playing, the display follows the latest voice's modulated WT position; with no active voice it follows the knob. Drag horizontally to select a frame, or wheel one frame at a time. The header's AUTO menu controls import cycle length. Auto reads Serum WAV `clm ` metadata when present, otherwise uses 2048 samples per cycle. An untagged power-of-two file of 128–2048 samples becomes one complete cycle. Manual lengths are 128, 256, 512, 1024, 2048 or 4096 samples; a manual choice takes precedence over metadata. Use 4096 manually for a single 4096-sample cycle.

The factory bank contains 128 waves ranging from sine to harmonic-rich shapes. Imported tables use the left channel, preserve cycle amplitudes, and remove DC. Up to 256 complete cycles are extracted; trailing partial cycles are ignored, except files shorter than one selected cycle, which are zero-padded. For ordinary recordings use Sampler or Granular, or choose the intended cycle length manually. Frame length and imported audio are embedded in saved presets and host state.

Sampler plays the imported audio at the selected ROOT MIDI note (default 60). START and END choose a region; LOOP repeats it with a short crossfade; REVERSE reverses playback. Without LOOP, playback ends at the selected boundary. Root sets the reference pitch, not automatic pitch detection. Stereo files retain both channels; mono files feed both channels. Unison and pitch bend still apply.

Sampler / PLAYBACK selects STRETCH and INTERPOLATION. Off is conventional pitched playback. Tape links speed and pitch: a 2× TIME FACTOR doubles duration and halves pitch. Granular and WSOLA separate sample traversal from note pitch, using overlapping 40 ms windows. WSOLA aligns windows by waveform correlation for smoother periodic

material; Granular keeps a more textured overlap. TIME FACTOR ranges from 0.125× to 8× (2× means twice as long). These are time-domain algorithms: extreme factors and transients can have audible window artefacts.

Interpolation choices are Linear, Cubic, Sinc 16 and Sinc 32. Sinc modes use a windowed band-limited resampler with anti-alias filtering when reading faster than source rate; Sinc 32 uses more CPU. Stretch/quality switches crossfade over 20 ms. Stretch is independent per sampler and does not expose Granulizer controls.

Granular offers two models. **Granulizer** is the default for new patches; **Cloud** retains the eight Hann-windowed grains, size, density, scan and spray controls from 0.4. Existing projects select Cloud automatically.

In Granulizer, GRAINS contains ATTACK, HOLD, GRAIN SPACE and WAVE SPACE. ATTACK sets both fade-in and fade-out, so total grain duration is twice ATTACK plus HOLD. GRAIN SPACE is the interval between grain starts relative to HOLD; WAVE SPACE advances the source by HOLD times that percentage at every start. With zero ATTACK and both spacings at 100%, playback reconstructs the sample at its original pitch. Increasing GRAIN SPACE slows traversal without changing pitch. WAVE SPACE can be negative: this reverses the order of grains while each grain still reads forwards. REVERSE separately reverses each grain's samples. Up to 64 grains overlap per unison voice; further starts are skipped if full.

MOTION contains alternating grain PAN, wave-spacing modulation DEPTH and SPEED, RANDOM source-position variation, TIME HOLD (freezes traversal), LOOP and REVERSE. TIME contains transient mode Off / Regions / Detect, transient HOLD, KEY TO, TEMPO SYNC and BEATS. Regions uses imported WAV cue points; Detect uses a local onset detector. A crossed marker aligns the grain and replaces its hold time. KEY TO selects Pitch, Percent, Step or Transient. Pitch follows played notes; the other modes keep source pitch and use keys relative to ROOT for position: +12 semitones selects 50% of the sample, advances 12 holds, or selects region 12, respectively. Negative position offsets clamp to the first position/region. TEMPO SYNC fits traversal to BEATS at the current tempo without repitching grains. START sets the initial location; automation moves the active scanner. LOOP wraps over the sample to END; TIME HOLD keeps creating grains at the current location.

Click an oscillator's waveform strip or select SAMPLE in the rack for the larger waveform view. Choose A1/A2/B1/B2. Drop a WAV/AIFF/FLAC onto the waveform or its header to replace that selected slot. The slot selection is independent of the Part A/B editing tab. The existing frame-size choice is used. A Classic slot switches to Sampler after a successful import; Wavetable/Sampler/Granular retain their mode. An invalid file keeps the previous source and mode. While dragging, the waveform highlights the destination. Loading and waveform-peak generation run in the background; long samples keep waveform zoom/pan responsive. It draws the actual stereo amplitude envelope, region markers, start/end and the most recent active voice's scanner, plus up to 16 active grain positions. Click to change START; wheel zooms around the cursor; Shift-wheel pans. The display is a view of playback, not a sample editor.

Enable OSC ENVELOPE in ENV to give a slot an additional attack, decay, sustain and release. A part's shared envelope still limits its final output. New WT Pos / Start A1/A2/B1/B2 targets individual players; A/B/A+B target groups. These destinations scan wavetable frames or sample locations. Grain size A/B/A+B scales grain duration. Position changes are smoothed.

IMPORT accepts WAV, AIFF and FLAC at 8–96 kHz, without a fixed duration limit. The four imported sources share a 512 MiB decoded-stereo-PCM budget. Temporary decoding, wavetable, waveform-cache and state-saving buffers use additional RAM. Alternatively, drop one file onto an oscillator panel: a Classic slot switches to Sampler, while the other engine selections are retained. Import decoding and table preparation run on a worker, and the footer reports completion or errors. RESET returns the slot's material to the factory glass sample/table. The waveform strip previews the loaded material. No original media path is needed on project recall: imported PCM data is compressed into the plug-in state, with a 1 GiB serialized-state limit. Save larger libraries separately; this is a single-sample source rather than a multisample instrument.

4. Filters and shared Filter Control

Each part has an input selector, type, slope, cutoff, resonance and drive. Input A uses A1+A2, B uses B1+B2, A+B uses both, and None disconnects the input. Consequently, a part's filter can process the other oscillator source. Its output still passes through its own amplitude envelope and Mix level.

Choose low-pass, high-pass, band-pass or bypass. Slope selects 12 or 24 dB per octave. Cutoff is the local frequency. Resonance emphasises the cutoff region; high resonance can self-oscillate while a note is active. Drive adds saturation.

The shared FILTER CONTROL affects both A and B while retaining their local settings. CUTOFF is an offset in semitones: +12 doubles both cutoff frequencies, -12 halves them. RESO adds a shared resonance offset. KEYTRACK ties cutoff to played pitch; at full tracking, an octave higher doubles cutoff. WARM changes the filter saturation characteristic. There is no third filter in this section. Modulation is combined with these settings before the effective cutoff is limited to a valid frequency for the current sample rate.

5. Modulation

Mod Env 1 and 2 each provide an ADSR envelope and two destination/amount pairs. LFO 1 and 2 also provide two pairs each. Misc 1 and Misc 2 each contain two source blocks, with two pairs per block. There are 16 routes in total. The source menu in a Misc block chooses the signal used by both routes in that block.

Choose a destination, then move its small amount knob away from zero. Amount ranges from -10 to +10; negative values reverse the modulation. None or amount zero makes a route inactive. Routes to the same destination are added together. The destination popup groups oscillators, filters and modulation/effects into three columns. The Source position / grains section contains WT Pos / Start and grain-size destinations.

Destinations include Volume, Pitch, Phase and Pan for A, B or A+B; Cutoff and Resonance for A, B or A+B; Mix A, B or A+B; Phaser Frequency; LFO 1/2 Rate and Gain; Distortion Amount; WT Pos / Start; and Grain size. Modulating A+B affects both parts together.

Misc sources include Velocity, Keytrack, Aftertouch, Mod Wheel, Pitch Wheel, Amp Env A/B, Mod Env 1/2, LFO 1/2, Step Velocity and random sources. Polyphonic pressure takes precedence over channel pressure for the affected note. Random values are generated at note-on. Step Velocity follows the active arp step and returns to zero after its gate ends.

Each LFO has waveform, Rate, Gain, Offset and FREE. With SYNC enabled and FREE off, RATE is a rhythmic division at the host tempo. With FREE enabled it uses its Hz rate and does not restart on each note. Otherwise notes restart the LFO. Gain scales its output; Offset shifts it. The global fallback tempo is 120 BPM when no host tempo is available.

To modulate wavetable position, set the oscillator to Wavetable and place WT POS near 50%. In LFO 1 or LFO 2 choose **WT Pos / Start A1**, **A2**, **B1**, **B2**, **A**, **B** or **A+B**, then raise its amount knob, for example to +5. Individual targets A1, A2, B1 and B2 affect only that player. A controls A1/A2, B controls B1/B2 and A+B controls all four. Individual and grouped routes can be combined; their offsets add before the result is limited to the bank range. Both destination/amount pairs of either LFO support this route. Set LFO Gain above zero, then choose a waveform and a free Hz rate or synced division. Positive and negative amounts scan in opposite directions; a centred WT POS allows a bipolar LFO to move both ways without clipping at the bank edges. The central wavetable view follows the modulated frame; the WT POS knob retains its base position. With Sampler or Granular this same destination modulates the sample start/scanner position. Use a bank with multiple distinct frames to hear wavetable scanning; a single-cycle, one-frame file has no other frame to select.

FM Index, AM Depth and Pulse Width targets each offer A1, A2, B1, B2, A, B and A+B. Individual and grouped routes add. At amount +10 and source +1 the offsets are +8 FM index, +1 AM depth and +1 pulse width, limited to valid ranges. Both MOD ENV blocks start with A/D/S/R and both route amounts/destinations at zero on a fresh instance. Set sustain or envelope times and a route to use them.

Modulation depth is specific to logisth. At amount +10 and source value +1, pitch adds 48 semitones, cutoff adds 120 semitones, phase adds one cycle, and pan adds one full side. Volume, Mix, Resonance, Gain and Distortion add one linear unit. LFO Rate multiplies by 16 and Phaser Frequency by 1024. Smaller amounts give smaller changes; start low. Effects are global, so FX modulation uses the newest active voice. LFO feedback modulation uses the previous sample.

6. Chords and pattern arpeggiator

The CHORDS rack page and Pattern arp use a chord-programming and multi-lane workflow inspired by Cthulhu, with logisth's own interface and implementation. It is not a Cthulhu clone and does not import Cthulhu chord/pattern files. The chain is MIDI input □ Chords □ Arpeggiator □ Synth and optional MIDI output. Each processor has its own enable square in the rack.

Chord mapper

Choose CHORDS and enable its square. TRIGGER selects one of 128 MIDI notes. VOICING assigns Unison, Major, Minor, Maj 7, Min 7, Dominant 7, Sus 2, Sus 4, Power, Diminished or Augmented to that trigger. MAP ALL assigns the selected type to all triggers, transposed to each root. Eight note fields edit the stored voicing directly: drag or enter MIDI numbers 0–127; -1 removes a voice. Empty triggers pass their original note through. Shared notes in overlapping chords are reference-counted so releasing one trigger does not cut another chord short.

LEARN collects up to eight incoming MIDI notes for the selected trigger. Play your voicing, then choose DONE. Notes are collected silently during learning; DONE stores the first eight distinct pitches in ascending order. The complete chord map is saved with the project. Chords can play as blocks with arp disabled, or supply a pool of notes to the Pattern engine.

Available arpeggiator modes

The engine menu contains **Legacy** and **Pattern**. The experimental Graphs mode has been removed from the interface, audio/MIDI scheduler and host parameter list. It cannot be enabled by an old Graphs setting. Twelve A–L pattern memories remain; switching slots automatically retains edits. SAVE explicitly stores the selected slot, and RAND randomizes the Pattern grid. Presets embed the entire memory bank.

MIDI OUT sends generated notes and supported pitch/controller/pressure messages to the host. MIDI ONLY also silences synth output; turning it on enables MIDI OUT. Route output in the DAW or choose an output device in Standalone settings. Turning MIDI OUT off sends all-notes-off. MIDI routing depends on the host. Large SysEx messages are not forwarded.

Pattern engine

Choose Pattern to use the seven-row grid, with note rank, octave, velocity, gate, repeat, chance and tie rows, independent lane lengths and two 16-step pages. Draw or right-click for values. Existing 0.4 projects select this engine automatically. Its pattern selector also uses the A–L memory bank. Legacy remains a separate mode with its original controls.

Legacy mode and compatibility

Legacy retains the ten original modes, velocity modes, 16 steps, transpose, wrap, gate, swing, octaves and hold. The Step sequence factory sound uses it. Older logisth states select Legacy automatically when they have no Pattern engine parameter. ARP enable clears held notes; play fresh notes afterward. Pattern uses note-rank/octave/gate lanes instead of Legacy transpose/wrap and its shared gate. Legacy controls remain available in the Legacy view.

7. Effects rack

Signal order is fixed: Distortion □ Phaser □ Chorus □ EQ □ Reverb □ Delay □ Compressor □ Main. Click a name to edit it and use the adjacent square to bypass or enable it. All seven effects are global after the A/B mix.

Section	Controls and use
Distortion	Overdrive, Foldback, Clip, Decimate and Bitcrush; Amount and Dry/Wet set character and blend.
Phaser	Six-stage stereo phasing with centre frequency, spread, L/R offset, width, LFO, feedback and Dry/Wet. Its frequency is a modulation destination.
Chorus	Delay, depth, rate, feedback, dual mode, width and Dry/Wet thicken a sound.

Section	Controls and use
Equalizer	Legacy two-shelf or 10-band Graphic; response curve, Auto Gain and output trim.
Reverb	Hall, Room, Spring, Vintage and Shimmer, with shared size/damping/predelay/width/mix and mode-specific controls.
Delay	Independent left/right times or divisions, feedback, ping-pong, spread, width, low/high cut, smear and Dry/Wet.
Compressor	Stereo-linked threshold, ratio, attack and release control dynamics.

Equalizer

MODE selects Legacy or Graphic. Legacy uses corrected bass/treble shelves with separate corner frequencies; a bass boost no longer applies an unrelated global attenuation. Graphic provides ten ISO bands at 31.5, 63, 125, 250, 500 Hz, 1, 2, 4, 8 and 16 kHz. Each band has ±18 dB gain. Move its fader or drag its point vertically in the response display; double-click resets that band to zero. The display shows the combined response including output gain/compensation.

OUT adds ±18 dB after either EQ. AUTO GAIN is off by default: enabling it reduces gain by the maximum positive response estimated across the audible spectrum. It is peak-response compensation, not a loudness meter or limiter; boosts, multiple voices and the following effects can still overload the output. Band/frequency changes and Legacy/Graphic transitions are smoothed over about 15 ms. Both modes keep their separate settings in presets and host state.

Reverb types

TYPE selects five algorithms. All now use denser 16-line feedback delay networks with stereo input diffusion (eight stages per channel, four for Spring). Hall is rebuilt for a spacious diffuse tail. Room adds early reflections; Spring uses frequency-dependent all-pass dispersion. Vintage adds dark filtering, saturated feedback and modulated delay lengths. Shimmer uses four overlapping pitch-shift windows in its excitation and feedback for rising/falling harmonics.

Size, Damp, Predelay (0–200 ms), Width, Dry/Wet and Diffusion are shared. Diffusion controls the input all-pass scattering. All five modes expose Decay (0.2–20 seconds, nominal; damping changes frequency-dependent decay). Spring adds Dispersion. Vintage adds Wow / Flutter. Shimmer adds Shimmer amount and Pitch (-12 to +24 semitones); +12 shifts up an octave, +7 adds a fifth and -12 shifts down an octave. Pitch-shifted tails are intentionally diffuse. Controls irrelevant to the selected type are hidden.

Type changes crossfade over approximately 100 ms, with storage prepared before playback. Presets and DAW state save the type and all model settings, including those currently hidden. Main calibration is unchanged.

When SYNC is on, rate/time fields that support synchronization become division menus. Turning SYNC off exposes their independent Hz/time values again. Bypass can freeze a section's internal memory; a tail may resume when re-enabled. Delay memory is limited to 16 seconds per channel, including very slow synced divisions. Compressor attack/release and some effect delays use milliseconds; envelope times use seconds internally. The panel displays suitable units.

8. Mixer and performance

The mixer scope has A, B and L+R source buttons. A/B show each part's stereo sum after its filter, amplitude envelope and Mix level, before shared effects and Solo, scaled by MAIN. L+R shows the actual final left-plus-right output, after effects, MAIN and MIDI ONLY. It is a sum, not an average. The scope uses zero-crossing alignment and automatic visual scaling; the peak label gives the unscaled peak. Selecting a source changes only the view and is saved in state.

Output calibration (0.6.0)

One centred Classic sine, one unison voice, MIDI note 69 at velocity 127, source Volume / Mix / Main at 100%, amplitude sustain at 100%, bypassed filters and effects, and no modulation produces **0 dBFS peak on each output channel**. The

unweighted RMS is -3.0103 dBFS (a full-scale sine has RMS 1/sqrt(2)). This is a unity reference, rather than a normal listening preset.

The source bus is calibrated before the shared FX. The previous 0.15 attenuator and the duplicate engine Master ramp have been removed. Main is applied once, after effects. Reset starts its smoothing at the chosen Main value. Each voice's amplitude envelope has its own approximately 1 ms anti-click edge; there is no extra startup fade on Main. Knob changes remain smoothed. Distortion and compressor thresholds therefore respond to the actual source-bus level. Levels and effect drive have been retuned together; older patches can sound different. Factory Main trims provide headroom; new instances start with Main at 65%. Reduce Main for louder layered edits.

The test reference for factory presets is a six-second C3/G3/C4/E4 chord (MIDI 48/55/60/64), velocity 102, including the attack. At 48 kHz most factory presets have individual reference levels; the rebuilt Hall and corrected EQ change some wet-preset peaks. Grain cloud and Chord pulse remain quieter. All ten are tested at 44.1, 48 and 96 kHz. The nine authored sounds keep at least 3 dB of reference-chord headroom; Init / saw at the requested Main 65% stays below full scale but has less than 3 dB of headroom. This reference does not guarantee headroom for every possible note or edit.

The mixer prints the peak of the latest processed block in dBFS. Values above 0 dBFS turn red. Constant-power panning is retained: hard panning raises the active channel by 3.01 dB relative to centre. Two identical oscillators add 6.02 dB; eight coherent, undetuned unison voices add 9.03 dB. Keep headroom with Mix/Main when stacking voices. There is no automatic note-count normalization or hidden output limiter. The scope's L+R mode is a literal sum and reads +6.02 dB relative to either channel for a centred mono sine.

The release calibration ZIP includes a full-scale 440 Hz float WAV, a rendered logisth WAV, a .logisth calibration preset and before/after CSV readings. Load the preset, play MIDI 69 at full velocity, leave the DAW track fader at unity, and bypass DAW effects to repeat the reference. Some RMS meters add an AES17 3 dB reference offset; these may display 0 dB RMS for this same signal.

MIX A and MIX B set the part levels; MAIN controls output after all effects. Independent L/R meters show post-effects output. There is no hard output limiter: loud patches, resonance and effects may exceed 0 dBFS. Reduce MAIN as needed.

POLY sets up to 16 simultaneous notes. Each note may contain several unison voices. SOLO isolates A or B. MONO / LEGATO uses last-note priority and returns to the previous held note when the newest one is released. POLY is disabled in mono mode. GLIDE sets portamento time; Normal mode glides for overlapping notes, while Slide also permits glide between separate notes.

BEND / st sets pitch-wheel range in semitones. The on-screen BEND wheel springs back to the centre after dragging. MOD remains where you set it. Route Mod Wheel through a Misc block to hear its effect. MIDI pitch bend and modulation messages also update these controls. The keyboard covers five octaves.

9. Presets, automation and project recall

The ten factory sounds are Init / saw, Wide supersaw, Sub bass, Slow atmosphere, Pulse pluck, Step sequence, Morph glass, Sample keys, Grain cloud and Chord pulse. The sound selector lists factory sounds and user presets from the current folder. Its arrows step through that list. Loading replaces the current patch.

The PATCH menu provides Save as, Save current, Load file, Choose folder, Refresh library and Delete current. Save as creates a named **.logisth** file; Save current overwrites the active user preset. Delete asks for confirmation and moves the file to the OS trash. The default library is logisth/Presets under the OS application-data directory. Choose folder selects an instance's working library; it does not relocate files. Load file can open a preset from elsewhere. Save/load run on a worker, with completion/errors in the footer. Wait for the completion message before closing the instance or moving a newly saved file.

Presets contain all 635 audio parameters, routing, compressed audio for all four slots, WAV regions, chord mappings, twelve named patterns, the preset comment and scope selection. Theme and editor size are separate user preferences and are retained when sounds change. The original sample files are unnecessary on recall. Saving uses a temporary file and atomic replacement. Invalid files leave the current patch intact. State/preset files are limited to 1 GiB. These are

logisth's own files; renaming another format to .logisth does not convert it.

All 635 parameters are available for host automation. The first 514 parameter IDs and order from 0.4 are retained. Granulizer and global MIDI-only controls follow them; the former Graphs parameter block is removed. A/B views, rack pages, theme and scope source do not change the patch's audio settings. DAW state contains the same embedded material and pattern data as a user preset. Restoring older states automatically selects the corresponding earlier granular/arp engines. Destination menus now contain 59 entries, including individual and grouped WT position, FM index, AM depth and pulse-width targets. Numeric destination IDs are retained, but normalized host automation may need review after a destination-list expansion.

There is no Sylenth1 bank import or MIDI Learn menu. Legacy prototype states receive a best-effort migration of filter and modulation settings; changes to DSP mean that identical sound is not guaranteed.

10. Sound-design examples

For a wide lead, load Wide supersaw. Raise unison detune gradually, open shared CUTOFF, and add a little Delay. Use MONO / LEGATO with a short GLIDE for connected melodies. Lower MAIN if the combined voices become too loud.

For an envelope sweep, load Init / saw, lower Filter A cutoff, select Cutoff A as Mod Env 1's first destination, and add a small positive amount. Set attack near zero, moderate decay, low sustain and short release. Increase the amount until the filter opens sufficiently at the start of each note.

For vibrato, choose Mod Wheel in a Misc source block and route it to LFO 1 Gain. Set LFO 1 to a sine wave in FREE mode at a few Hz, set its base Gain to zero, then route LFO 1 to Pitch A+B with a small amount. The wheel opens the vibrato depth. Keep amounts low to avoid large pitch excursions.

For a sequence, load Step sequence and hold a note. Adjust PITCH and VELOCITY, set a zero velocity for a rest, and enable HOLD to tie steps. Select 09–16 and increase LENGTH to use the second half. Automate the shared filter cutoff to move both parts together without replacing their local settings.

11. Troubleshooting and technical notes

High pitches in Sampler/Granular use interpolated resampling and may alias. Wavetable mip bands reduce aliasing, but phase warp can introduce new harmonics. These engines are independent implementations, not models of commercial synths. Dense granular patches with high unison/polyphony require more CPU. Imported material can be loud; use moderate levels. An import failure leaves the previous material intact. A source starts again on the next note or engine/material change.

Version 0.7 rebuilds wavetable frame selection and visualization and removes the experimental Graphs arp. Legacy, Pattern and the Granulizer remain available. Native release archives are published only after their platform builds and tests pass. See BUILD_INFO.json in each archive for source revision and validation results.

No sound: check the audio/MIDI device or DAW routing, oscillator Voices and Volume, filter input, envelope sustain, Mix levels and MAIN. Check SOLO and whether the arp needs fresh notes. A long attack or low cutoff can hide a note.

No modulation: check the source is active, destination is not None, and amount is nonzero. The mod wheel needs an assigned route. A source can be zero during part of a note; Step Velocity needs an active arp gate.

Missing control: A/B selects the part, a rack name selects effect controls, and SYNC/FREE selects time versus rhythmic controls. PHASE works independently of RETRIG in Classic and Wavetable. The second group of arp steps is under 09–16.

High CPU: reduce unison voices, polyphony or active effects. Some abrupt changes to routing, waveforms and delay times can click. Selected continuous controls are smoothed, but not every transition is crossfaded.

Build validation includes audio at 44.1/48/96 kHz, filters, modulation, arp timing, state restoration, preset output and editor bindings/layout/theme persistence. Linux tests additionally load the actual CLAP and VST3 binaries. A VST3 factory smoke test does not replace testing in your particular DAW. Native macOS tests run on Intel; Universal

architecture checks establish that arm64 code is present, not that it has been auditioned on Apple Silicon hardware.

The filter is an independent SVF design. Hall uses JUCE Reverb; the other four reverb algorithms are original wavelogik implementations. Modulation scales and some sequencer behaviour are implementation choices. No sample-exact comparison with Sylenth1 has been performed. Third-party font notices are included in assets/fonts. The original Sylenth1 manual and graphics are not redistributed.

License and manufacturer

Manufacturer: wavelogik. Product: logisth. Original code and documentation are Copyright (c) 2026 wavelogik, licensed under MIT. Libraries, SDKs and fonts retain their upstream licenses; see LICENSE and THIRD_PARTY_NOTICES.md. The current plug-in identifiers use com.wavelogik.logisth and manufacturer code Wvlg. This changes the host instrument identity from earlier builds; .logisth sound files remain readable.